Tile Painter

Time limit: 2 seconds | Memory limit: 256 MB

Tilly the tile painter decorates long corridors of the Codeforcia palace. Every corridor is covered by a strip of square tiles, and each tile is painted with one lowercase Latin letter. The pattern of a strip is therefore described by a string `s`.

Tilly is lazy, so she wants to paint as little as possible. She looks for the shortest string `t` such that the whole strip can be obtained by writing `t` one after another some number of times (at least once). In other words, `s = t + t + … + t`, where "+" means concatenation.

For example, the strip `abcabc` can be made from the tile pattern `abc` written twice, while the strip `abcab` cannot be made by repeating any string other than itself, so its pattern is the whole strip `abcab`.

Write a program that finds the length of the shortest such string `t` for each strip.

Input

The first line of input contains a single integer `T` ($1 \le T \le 10^4$) representing the number of test cases. This is followed by a description of each test case.

Each test case consists of a single line containing a string `s` ($1 \le |s| \le 2 \cdot 10^5$) consisting of lowercase Latin letters.

It is guaranteed that the sum of $|s|$ across all the test cases will not exceed $2 \cdot 10^5$.

Output

For each test case, output a single integer: the length of the shortest string `t` such that `s` is equal to `t` repeated one or more times.

Examples

Input	Output
6 abab aaaa abc abcabcabc abaaba ababa	2 1 3 3 3 5

Note

In the first test case, `abab = ab + ab`. The string `a` does not work, so the answer is 2.

In the second test case, `aaaa` is the letter `a` written four times, so the answer is 1.

In the third test case, all letters are different, so the only possible pattern is the whole strip and the answer is 3.

In the fourth test case, `abcabcabc = abc + abc + abc`.

In the fifth test case, `abaaba = aba + aba`. Shorter strings such as `ab` do not work, since `ab + ab + ab = ababab`.

In the sixth test case, the first two letters `ab` repeat, but the length 5 is not divisible by 2, so the strip cannot be split into equal copies. The only pattern is the whole strip, and the answer is 5.